

```
In[*]:= << mGRG`STensor`
```

Chap. 6 Differential Form Applications

6.1 The Theorem of Stokes

```
In[*]:= Off[XDtoCDfrag]
```

2차원 시공간:

```
In[*]:= SetDimension[2]
```

```
In[*]:= Fdefine[w1, 1]
```

```
In[*]:= w1  
CoordRep[w1, {x, y}]
```

Out[*]=

w1

Out[*]=

$w1_1 dx + w1_2 dy$

```
In[*]:= XD[w1]  
CoordRep[%, {x, y}]
```

Out[*]=

dw1

Out[*]=

$(-\partial_2 w1_1 + \partial_1 w1_2) dx \wedge dy$

3차원 시공간:

```
In[*]:= SetDimension[3]
```

```
In[*]:= w1  
CoordRep[w1, {x, y, z}]
```

Out[*]=

w1

Out[*]=

$w1_1 dx + w1_2 dy + w1_3 dz$

```
In[*]:= XD[w1]
CoordRep[%, {x, y, z}]
```

```
Out[*]=
dw1
```

```
Out[*]=

$$\left(-\partial_2 w_{11} + \partial_1 w_{12}\right) dx \wedge dy + \left(-\partial_3 w_{11} + \partial_1 w_{13}\right) dx \wedge dz + \left(-\partial_3 w_{12} + \partial_2 w_{13}\right) dy \wedge dz$$

```

```
In[*]:= Fdefine[w2, 2]
```

```
In[*]:= w2
CoordRep[w2, {x, y, z}]
```

```
Out[*]=
w2
```

```
Out[*]=

$$w_{212} dx \wedge dy + w_{213} dx \wedge dz + w_{223} dy \wedge dz$$

```

```
In[*]:= XD[w2]
CoordRep[%, {x, y, z}]
```

```
Out[*]=
dw2
```

```
Out[*]=

$$\left(\partial_3 w_{212} - \partial_2 w_{213} + \partial_1 w_{223}\right) dx \wedge dy \wedge dz$$

```

```
In[*]:= On[XDtoCDfrag]
```

6.2 Differential Forms and Maxwell's Equations

IndexedTensor를 이용한 Maxwell 방정식1:

```
In[*]:= Tdefine[F, "2-"]; Tdefine[J, 1]
```

```
In[*]:= maxwellEq = CD[ub, F[la, lb]] -  $\frac{1}{c^2}$  J[la]
```

```
Out[*]=

$$\nabla^b F_{ab} - \frac{J_a}{c^2}$$

```

미분 형식을 이용한 Maxwell 방정식1:

```
In[*]:= Fdefine[f, 2]; Fdefine[j, 1]
```

```
In[*]:= CoXD[f]
FtoC[%, {la}]
```

```
Out[*]=
```

$$\delta f$$

```
Out[*]=
```

$$-\nabla^b f_{ba}$$

```
In[*]:= eqdiv = CoXD[f] - \frac{1}{c^2} j
FtoC[%, {la}]
```

```
Out[*]=
```

$$-\frac{j}{c^2} + \delta f$$

```
Out[*]=
```

$$-\nabla^b f_{ba} - \frac{j_a}{c^2}$$

IndexedTensor를 이용한 Maxwell 방정식2 (비앙키 항등식):

```
In[*]:= cycEq = 3 * AntisymmetrizeIndices[CD[la, F[lb, lc]], {la, lb, lc}]
% // TindexSort
```

```
Out[*]=
```

$$\frac{1}{2} \nabla_a F_{bc} - \frac{1}{2} \nabla_a F_{cb} - \frac{1}{2} \nabla_b F_{ac} + \frac{1}{2} \nabla_b F_{ca} + \frac{1}{2} \nabla_c F_{ab} - \frac{1}{2} \nabla_c F_{ba}$$

```
Out[*]=
```

$$\nabla_a F_{bc} - \nabla_b F_{ac} + \nabla_c F_{ab}$$

미분 형식을 이용한 Maxwell 방정식2 (비앙키 항등식):

```
In[*]:= XD[f]
FtoC[%, {la, lb, lc}]
```

```
Out[*]=
```

$$df$$

```
Out[*]=
```

$$\nabla_a f_{bc} - \nabla_b f_{ac} + \nabla_c f_{ab}$$

벡터 퍼텐셜 1-form과 전자기장 2-form:

```
In[*]:= Fdefine[a, 1]
```

In[*]:= **vecpotEq = f - XD[a]**
FtoC[%, {la, lb}]

Out[*]=

$$\mathbf{f} - \mathbf{da}$$

Out[*]=

$$-\nabla_a a_b + \nabla_b a_a + f_{ab}$$

벡터 퍼텐셜 1-form을 이용한 Maxwell 방정식2 (비앙키 항등식):

In[*]:= **XD[vecpotEq]**
FtoC[%, {la, lb, lc}]

Out[*]=

$$\mathbf{df}$$

Out[*]=

$$\nabla_a f_{bc} - \nabla_b f_{ac} + \nabla_c f_{ab}$$

벡터 퍼텐셜 1-form을 이용한 Maxwell 방정식1:

In[*]:= **eqdiv2 = eqdiv /. f -> XD[a]**

Out[*]=

$$-\frac{\mathbf{j}}{c^2} + \delta \mathbf{da}$$

로렌츠 게이지 조건:

In[*]:= **lorentz = CoXD[a]**
FtoC[lorentz, {}]

Out[*]=

$$\delta \mathbf{a}$$

Out[*]=

$$-\nabla^a a_a$$

로렌츠 게이지 조건을 요구하였을 때의 Maxwell 방정식1:

In[*]:= **CoXD[XD[a]] + XD[CoXD[a]] - $\frac{1}{c^2} \mathbf{j}$**
FtoC[%, {la}]

Out[*]=

$$-\frac{\mathbf{j}}{c^2} + \delta \mathbf{da} + \mathbf{d}\delta \mathbf{a}$$

Out[*]=

$$-\nabla_a \nabla_b a^b + \nabla^b \nabla_a a_b - \nabla^b \nabla_b a_a - \frac{j_a}{c^2}$$

In[*]:= **maxdivEq = %**

Out[*]=

$$-\nabla_a \nabla_b a^b + \nabla^b \nabla_a a_b - \nabla^b \nabla_b a_a - \frac{j_a}{c^2}$$

In[*]:= **maxdivEq[[1]] = CommuteCD[{la, lb}, maxdivEq[[1]]]**

Out[*]=

$$-\nabla_b \nabla_a a^b + a^b R_{ab}$$

따라서 로렌츠 게이지 조건을 요구하였을 때

In[*]:= **maxdivEq // Tsimplify**

Out[*]=

$$-\nabla_b \nabla^b a_a - \frac{j_a}{c^2} + a_b R_a^b$$

6.3 Tensor-valued Forms: Cartan Structure Equations

DiffForm (basis 1-form, curvature 2-form, connection 1-form, torsion 2-form)을 정의한다:

In[*]:= **Fdefine[e[ua], 1]; Fdefine[r[la, ub], 2];
Fdefine[ω[la, ub], 1];
Fdefine[t[ua], 2]**

In[*]:= **e[ua] ^ e[ub]
FtoC[%, {lm, ln}]**

Out[*]=

$$e^a \wedge e^b$$

Out[*]=

$$-e_m^b e_n^a + e_m^a e_n^b$$

In[*]:= **r[la, ub]
FtoC[%, {lm, ln}]**

Out[*]=

$$r_a^b$$

Out[*]=

$$r_{mna}^b$$

Cartan 방정식1:

$$t^a = de^a - e^b \wedge \omega_p^a$$

In[*]:= **cartan1 = XD[e[ua]] - e[ub] ^ ω[lb, ua] - t[ua]**

Out[*]=

$$-t^a + de^a - e^b \wedge \omega_b^a$$

```
In[*]:= cartan1Rule = RuleUnique[XD[e[ua_]], t[ua] + e[ub] ^ ω[lb, ua]]
Out[*]=
```

```
deua := DumFresh[ta + eLatin$10900 ^ ωLatin$10900a]
```

Cartan 방정식2:

$$r_a^b = -(d\omega_a^b - \omega_a^p \wedge \omega_p^b)$$

```
In[*]:= cartan2 = XD[ω[la, ub]] - ω[la, uc] ^ ω[lc, ub] + r[la, ub]
Out[*]=
```

```
rab + dωab - ωac ^ ωcb
```

```
In[*]:= cartan2Rule = RuleUnique[XD[ω[la_, ub_]], -r[la, ub] + ω[la, uc] ^ ω[lc, ub]]
Out[*]=
```

```
dωla_ub := DumFresh[-rab + ωaLatin$10915 ^ ωLatin$10915b]
```

```
In[*]:= cartanRule = {cartan1Rule, cartan2Rule}
Out[*]=
```

```
{deua := DumFresh[ta + eLatin$10900 ^ ωLatin$10900a],
 dωla_ub := DumFresh[-rab + ωaLatin$10915 ^ ωLatin$10915b] }
```

미분 형식으로 표현된 비양키 항등식 1:

$$dt^a + t^p \wedge \omega_p^a = -e^p \wedge r_p^a$$

```
In[*]:= XD[cartan1]
% /. cartanRule
FtoC[%, {lp, lq, lr}]
```

```
Out[*]= -dta + eb ^ dωba - ωba ^ deb
```

```
Out[*]= -dta - eb ^ rba - ωba ^ tb
```

```
Out[*]= -∇ptqra + ∇qtpra - ∇rtpqa - erb rpqba + eqb rprba - epb rqrba - tqrb ωpba + tprb ωqba - tpqb ωrba
```

torsion-free일 때에는 $r_{[abc]}^d = 0$ 이 된다.

미분 형식으로 표현된 비양키 항등식 2:

$$dr_a^b - \omega_a^p \wedge r_p^b + r_a^p \wedge \omega_p^b = 0$$

```
In[*]:= XD[cartan2]
% /. cartanRule
```

Out[*]=

$$d\mathbf{r}_a^b + \omega_a^c \wedge d\omega_c^b - \omega_c^b \wedge d\omega_a^c$$

Out[*]=

$$d\mathbf{r}_a^b - \omega_a^c \wedge \mathbf{r}_c^b + \omega_c^b \wedge \mathbf{r}_a^c$$

```
In[*]:= bianchi = %;
```

DiffForm 표현과 IndexedTensor 표현의 비교:

```
In[*]:= Off[XDtoCDfrag]
```

```
In[*]:= bianchi
FtoC[%, {lc, ld, le}]
```

Out[*]=

$$d\mathbf{r}_a^b - \omega_a^c \wedge \mathbf{r}_c^b + \omega_c^b \wedge \mathbf{r}_a^c$$

Out[*]=

$$\partial_c r_{dea}^b - \partial_d r_{cea}^b + \partial_e r_{cda}^b - r_{def}^b \omega_{ca}^f + \\ r_{dea}^f \omega_{cf}^b + r_{cef}^b \omega_{da}^f - r_{cea}^f \omega_{df}^b - r_{cdf}^b \omega_{ea}^f + r_{cda}^f \omega_{ef}^b$$

torsion-free일 때에는 $\nabla_{[c} r_{de]a}^b = 0$ 이 된다.